

Defragmentation takes all the non-contiguous parts of a file and arranges them contiguously. It does this on a best-effort basis; if it can't make one contiguous stretch of the file, it will at least try and reduce the number of fragments.

The process itself takes any amount of time ranging from a few minutes to several hours, depending, of course, on how fragmented the disk is—and also on how much RAM you have and how fast your processor is. If you bought your computer say, six months ago, and have never defragmented it, chances are it will take several hours, and even then, it won't be able to do a perfect job. Make it a practice to defragment your disk(s) once a month or so. We recommend that you also defragment your disk after you've installed Windows and other essential software for the first time.

Apart from reducing wear and tear, there is another fundamental advantage to frequent defragmentation—performance. If your files are less fragmented, your hard disk will respond faster, leading to better performance.

9.1.1 Defragging Your Disk

There is more than one way to reach the disk defragmenter program from within Windows. The easiest is Start Menu > All Programs > Accessories > System Tools > Disk Defragmenter.

If you've used the defragmenter in Windows 98 or ME, you'll see that there are some changes. First, you don't get the option to "arrange files so that your programs start faster." Windows XP does this by default. Second, the old defragmenter would arrange files in such a manner that it appeared as though every single cluster was in place. We won't go into the internals of this, but XP does it differently—you'll find files arranged with free space in between them, for example.

The main defrag window in Windows XP looks something like this. At the bottom, there are two strips: the first one tells you what